Dinh (Kate) Lam

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Education

Harvey Mudd College | Dean's List *Expected May 2027*

B.S. in Engineering, Concentration in Economics, Emphasis in Data Science

- **Advanced coursework:** Dynamics of Rigid Bodies, Signals and Systems, Operations Research, Mech Design

Experience

Optimization Systems Analysis Intern at NASA Ames, AMA *June 2026 – present*

Moffett Field, CA

- Built and validated dynamic lookup aerotables integrating OpenVSP and CBAERO outputs to generate aerodynamic coefficients and stability derivatives for hypersonic entry vehicles.
- Extended Condor's spline Jacobian to compute derivatives with respect to aerodynamic coefficients, enabling gradient-based CobraMRV shape and trajectory optimization with results comparable to genetic algorithms.
- Reduced aerodynamic table-generation runtime by 75% through eliminating redundant spline construction.

Clinic Consultant at General Matter *August 2026 – present*

Hawthorne, CA

- Developing an IoT monitoring system for UF_6 cylinders integrating temperature, pressure, location, and tamper sensing to replace manual barcode-based tracking and reduce operator intervention.
- Designing low-power system architecture and iterating through custom PCB spins to integrate sensing, wireless communication, battery, and power-management subsystems.

Clinic Consultant at Lifemotion Medical Technology *Jan 2026 – May 2026*

Maastricht University, The Netherlands

- Designed, patented, and machined stabilizing components, dilator clamp, and lateral translation via lead screws for an automated ECMO cannulation device.
- Performed automated open-vessel cannulation during ex vivo sheep perfusion trial.
- Built MATLAB ultrasound segmentation for detecting vessel centerline and depth, enabling closed-loop needle and dilator positioning via potentiometers, lead screws, and multi-axes linear actuation.

Metallurgy Researcher and Engman Fellow in Applied Mechanics *Dec 2024 – May 2026*

University of New South Wales, Australia

- Achieved 4 at.% carbon solubility in copper-based lead-free high entropy alloys through iterative composition tuning, quantified via MATLAB image processing script of SEM, EDS, and optical microscopy images.
- Presented findings at the TMS Conference; resulting alloy is undergoing commercial development.

Undergraduate Researcher in Molecular Engineering Lab *Jan 2024 – Dec 2024*

Harvey Mudd College, CA

- Operated UV-Vis spectrometer and fluorometer to collect over 300 data points in 3 weeks on mixed micelles. Presented progress and findings on fluorescence properties of peptide amphiphiles at Soft Matter Symposium.

Makerspace Steward *Aug 2023 – present*

- Problem solve and guide Claremont consortium users on which tools to use to fabricate items and safely operate Makerspace machines including CNC, waterjet cutter, mill, lathe, welding station, laser cutter, 3D printers.

Projects

Van's Aircraft RV-12 and Experimental Electrical Vehicle *Aug 2025 - May 2026*

- Integrated avionics, canopy mechanisms, and latching components on RV-12 aircraft in 5-person build team.
- Built an EV kart combining battery technology and embedded software for motor control and system integration.

AUV Underwater Robots, Experimental Engineering *Aug 2024 - May 2025*

- Programmed autonomous underwater robot using MATLAB and Arduino; Integrated light, pressure, and pH sensors to measure conditions for aquatic photosynthesis, using real-time sensor feedback to regulate navigation.

Operations Research Hyperschedule Optimization Model *Sept 2025 - Dec 2025*

- Built a AMPL/Gurobi course-scheduling optimization model incorporating degree requirements, time preferences, travel distance, and accessibility constraints using linear/nonlinear optimization techniques.

Skills

Technical: MATLAB, Python, C++, SystemVerilog, JavaScript, SolidWorks, COMSOL, FEA, GD&T, CBAero, OpenVSP, Condor, AMPL, Gurobi, LabVIEW, Quartus, Fusion 360, Onshape, CNC, mill, lathe, waterjet, welding

Leadership: Class President, Dorm Residential Assistant; **Languages:** English, Vietnamese, Cantonese, French